

Lecture 10

Fundamental Subspaces of a matrix

① Column space $C(A)$

② Null space $N(A)$

③ Row space $C(A^T)$

④ Null space of A^T $N(A^T)$
↳ Left null space.

$A_{m \times n}$:

$N(A)$	in	$\mathbb{R}^n$
$C(A)$	in	$\mathbb{R}^m$
$C(A^T)$	in	$\mathbb{R}^n$
$N(A^T)$	in	$\mathbb{R}^m$

$\mathbb{R}^n$ $\rightarrow$ row space, null space

$\mathbb{R}^m$ $\rightarrow$ column space, left null space.

$$\dim(N(A)) = n - r \quad [\text{free columns are basis}]$$

$$\dim(C(A)) = r \quad [\text{pivot columns are basis}]$$

$$\dim(C(A^T)) = r$$

$$\dim(N(A^T)) = m - r \quad [\text{free cols in } A^T]$$

Doing row operations on A would change its column space.

$$\text{So, } C(R) \neq C(A)$$

• But their row space remains the same.

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

(r)

Basis $\rightarrow$ first 2 rows

$$\begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$$

First r rows of R is the basis of row space of A .

$$\underline{\underline{N(A^T)}}$$

$$R = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A^T \begin{pmatrix} y \end{pmatrix} = 0$$

$$\begin{pmatrix} y^T \\ \vdots \end{pmatrix} A = 0$$

$$y^T A = 0$$

↳ This is why left null space

$$y^T (E^{-1}R) = 0$$

$$A^T E^T = R^T$$

doing same ops on I_{max}

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

E



This vector produced the zero row.

So, basis of $N(A^T) = \lambda \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$

~ New vector space

All 3×3 matrices $= M$

$$A + B \in M$$

$$\lambda A \in M$$

$$aA + bB \in M \quad \forall a, b \in \mathbb{R}$$

eg subspace ~

- $[0]_{3 \times 3}$
- $\lambda I_{3 \times 3} \quad \lambda \in \mathbb{R}$
- $\lambda A_{3 \times 3}$
- Upper triangular
- Diagonal matrices
- Symmetric matrices.
- Lower triangular

$$\dim(\text{Diagonal}) = 3$$

$$\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \Rightarrow \underline{\underline{\text{Basis}}}$$